

PHASE 5 – PROJECT DEVELOPMENT

Development Environment

- Python virtual environment is used for isolated project dependencies.
- Dependencies are installed through requirements.txt.
- A .env file is used for the Gemini API key.

Backend Development

- FastAPI application provides the backend API.
- The backend includes a root/health verification flow and modular routes.
- The /generate endpoint accepts document information and returns generated legal content.

AI Integration

- The AI generation component forms a structured prompt from user inputs and sends it to the Gemini service.
- The generated text is returned to the FastAPI layer and displayed by the frontend.

Frontend Development

- Streamlit provides document-type, parties, terms and date inputs.
- The interface provides Generate Document, Edit Document and download functions.
- Generated content is shown in an editable preview.

Document Generation

- TXT provides plain editable text.
- DOCX uses python-docx for formatted Word documents, including titles, paragraphs and terms tables.
- PDF uses FPDF for formatted output with headings, terms and footer support.

Completed Workflow

- Input → API request → AI generation → preview → editing → export.